

Quiz 3

Name(s):

1. In this problem, you will prove the following statement:
Let $F : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a PRF. Prove that

$$G(s) = F_s(0^n) || F_s(1^n)$$

is a PRG.

The following questions are meant to guide you through the proof. If you feel that you do not need them, you can just provide the full proof at the end.

- (a) Write down the assumption you need to make to start the proof by reduction. (What do you need to assume about the adversary $\mathcal{A}_c$?)
- (b) In order to prove security by a reduction, what is the adversary $\mathcal{A}_r$ that you need to construct?
- (c) How would you construct $\mathcal{A}_r$ using $\mathcal{A}_c$?
- (d) Argue that $\mathcal{A}_r$ succeeds if $\mathcal{A}_c$ succeeds.

2. Let $F : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a PRF.
Is $F'_k(x) = F_k(x) || F_k(\bar{x})$ necessarily a PRF?

(a) What do you get when you query F'_k on input $x = 0^n$?

(b) Can you find two queries x, y such that $F'_k(x)$ and $F'_k(y)$ are correlated?

(c) How does this help you distinguish F'_k from a random function?